12 ab

End - End Communication at Transport Layer

a) Implement echo client server using TCP / UDP Socket

b) Implement a chat program using Socket program

TCP:

```
import socket
import threading
#--SERVER--
def start_server():
    server = socket.socket (socket.AF_INET, socket.SOCK_STREAM)
    server.bind (('127.0.0.1', 5000))
    server.listen (1)
    print ("[SERVER] Listening on 127.0.0.1:5000")
    conn, add = server.accept()
    printf (f"[SERVER] connected by {addr}")
    while true:
        data = conn.recv (1024).decode()
        if not data or data.lower() == 'quiet':
            print ("SERVER] connection closed by circuit")
            break
        print (f"(SERVER, connection closed by circuit")
    con.close()
    Server.close()

#-client --
    def start_client ():
        client = socket.socket (socket.AF_INET, socket.SOCK_STREAM)
        client.connect (('127.0.0.1, 5000)
        print ("[client] connected to server")
    While true:
        msg = input ("Enter message (quit / exit):")
        client.send (msg.encode)
```

```
if msg.lower() == 'quit':
        break
response = client.connect(127.0.0.1, 5000)
print(f"Server Response: {response}")

client.close()

# -- MAIN --
if __name__ == "__main__"
    server_thread = threading.Thread(target = start_server,
                                     daemon = True)

    server_thread.start()

    start_client()
```

Output:

```
[Server] Listening on 127.0.0.1 : 5000   [client]
connected to server
hello
[server] received hello
[Server Response] server received : hello
Enter message (or 'quit' or 'exit'):
```

UDP

```
import socket
import threading
# -- Server --
def udp_server():
    s = socket.socket(socket.AF_INET, socket.
    SOCKET_DGRAM)
    s.bind(127.0.0.1, 12000)
    print("[SERVER] Running on 127.0.0.1:12000")
```

```
while true :
    data, addr = s.recvfrom (1024)
    msg = data. decode()
    if msg . lower ()=='quit'
    print (f" [SERVER] client {addr} ended the
          - connection ")
    break
    print( "SERVER] Received from {addr} : {msg}" )
    s. sendto (data, addr)
s.close()
print ("ES ERVER]  STOPPED " )

#-- client --

def udp- client ():
    c= Socket. Socket (Socket . AF_INET, socket. SOCK_
                                        DGRAM)
    server = ('127.0.0.1'  , 12000)
    while true :
        msg = input (" Enter message (Or 'quit' to stop)
        c. send to (msg. encode(), server )
        if msg. lower() =='quit"
        print ("[CLIENT] closing connection ...") break
    data, _ = c .recvfrom (1024)
    print ( "Echo from server", data. decode())
c.close()
print ("[CLIENT] Stopped")

#. Main -

if __name__  =='__main__':
    threading . thread ( target = udp _server
```

demon= True) . start() udp.client()

Output:

Enter a message: [SERVER] Running on 127.0.0.1: 12000hi

[SERVER] Received from ('127.0.0.1', 49670): hi

Echo from server : hi

Enter message (or quit to stop) how are you

[SERVER]

Result:

Thus the end to end communication at transport layer is completed